LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 10 Test (Set B): The Human Eye and the Colourful World

Chapter / Topic: The Human Eye and the Colourful World

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Draw neat, labelled ray diagrams wherever required.

SECTION A

6 marks

Choose the correct option.

Q1. Which part of the eye controls the size of the pupil? **[1]**

(a) Cornea (b) Retina
(c) Iris (d) Eye lens

Q2. The splitting of white light into its component colours is called: **[1]**

(a) reflection (b) scattering
(c) refraction (d) dispersion

Q3. The clear sky looks blue because of: **[1]**

(a) reflection of light (b) scattering of light
(c) dispersion of light (d) atmospheric refraction

Q4. A person cannot see objects beyond 2 m clearly. The defect is: **[1]**

(a) myopia (b) presbyopia
(c) hypermetropia (d) cataract

Q5. Signals from the retina are carried to the brain by the: **[1]**

(a) ciliary muscles (b) optic nerve
(c) iris (d) cornea

Q6. **Assertion (A):** The sky appears dark to passengers flying at very high altitudes.
Reason (R): At such heights there is very little scattering of sunlight. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. What are the near point and the far point of a normal human eye? **[2]**

OR

What happens to the image distance in the eye when we move an object farther away? How does the eye adjust?

Q8. Why does the Sun appear white at noon but reddish at sunrise? **[2]**

SECTION C

6 marks

Q9. A person needs a lens of power −5.5 D to correct distant vision and a lens of power +1.5 D to correct near vision. **[3]**

(i) Find the focal length of the lens for distant vision. **[1]**

(ii) Find the focal length of the lens for near vision. **[1]**

(iii) What kind of spectacle lens would suit this person? **[1]**

Q10. Why do different colours bend through different angles in a prism? What is a spectrum? Write the order of its colours. **[3]**

OR

Why do stars twinkle while planets do not?

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Give the function of the cornea, iris, pupil, eye lens, ciliary muscles and retina. **[3]**

(ii) Why do we have two eyes rather than one? **[2]**

OR

(a) What is presbyopia? How do bifocal lenses help? [3]
(b) What is a cataract, and how is vision restored? [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. Eyes can live even after death! By donating our eyes after we die, we can light up the life of a blind person. The eyes must be removed within 4–6 hours of death. Many people in India suffer from corneal blindness, which can be cured by transplanting the cornea from donated eyes. One pair of eyes can give vision to up to four corneal-blind people. **[4]**

(i) Within how many hours after death must donated eyes be removed? **[1]**

(ii) Which part of a donated eye is used to cure corneal blindness? **[1]**

(iii) How many people can benefit from one pair of donated eyes? Give one reason why eye donation should be encouraged. **[2]**

~ All the best! ~